

1 LazyModeler: An R package for automatic 2 simplification, checking, and visualization of regression 3 models

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8 Summary

9 Setting up, simplifying, checking, and visualizing regression models continues to be a time-
10 consuming task involving multiple, sometimes concurrent, workflows and software packages.
11 This particularly applies to big data research where several models with different response
12 variables and many explanatory variables need to be set up and optimized. To tackle this
13 problem, we present LazyModeler - a statistical package for the programming language R
14 that allows users to easily perform regression modeling. It includes removal of autocorrelated
15 variables, choice between several types of (non)linear regression models, standard stepwise model
16 simplification, various model quality checks, plotting of coefficient estimates and relationships,
17 and output generation. LazyModeler will significantly speed up regression modeling, enabling
18 people to analyze and illustrate their data in a statistically reliable and standardized manner.

Statement of need

19 Statistical modeling describes the process of finding a mathematical function with specific
20 statistical assumptions that best fits the observed data ([Crawley, 2007, 2015](#); [Henley et al., 2020](#)). This process attempts, in practice, to find a (causal) relationship between a dependent
21 response variable y and an independent predictor variable x for any postulated hypothesis. For
22 statistical inference and graphics in science, the programming environment R ([R Core Team, 2026](#))
23 has become highly popular.

24 Linear models, as one of the most basic and powerful tools, have been frequently applied in
25 this context ([Crawley, 2007, 2015](#); [Li, 2023](#); [Schielzeth et al., 2020](#)). Because of their flexibility,
26 they also allow for non-normally distributed response variables (e.g., in the case of binomial,
27 proportional, or count data), and any kind of transformation for numerical (e.g., polynomial or
28 logarithmic) and categorical (e.g., centered or one-hot/fractional encoded) predictor variables,
29 as well as interactions among them ([Cai et al., 2023](#); [Henley et al., 2020](#); [Karbstein et al., 2019, 2020, 2021](#);
30 [Liaw et al., 2021](#); [Römermann et al., 2016](#); [Schielzeth, 2010](#)). Regression
31 models also provide the ability to control for random effects that may influence the variables of
32 interest (e.g., [Bauer & Albrecht, 2020](#); [Schielzeth et al., 2020](#); [Wicke et al., 2016](#)). Although
33 other statistical technologies can outperform them in highly complex, nonlinear scenarios,
34 regression models allow for detailed variable transformation and interaction, mathematical
35 formula specification, calculation of effect sizes, determination of variable significance, and
36 thus hypothesis testing and explanation ([Benjamin et al., 2018](#); [Bzdok & Ioannidis, 2019](#);
37 [Cai et al., 2023](#); [Karbstein et al., 2023](#); [Li, 2023](#); [Schulz et al., 2020](#)). Recent developments
38 make regression models also applicable to nonlinear scenarios (e.g., [Bates et al., 2024](#); [Hastie,](#)

2023). Consequently, they are of high practical value in finding and interpreting significant relationships.

In statistical modeling, and especially in real-world applications, multiple predictors are assumed for a given response variable. As a consequence, people strive to exclude the irrelevant from the relevant (statistically significant) information, which is called model simplification (Crawley, 2007, 2015; Forstmeier & Schielzeth, 2011). One of the most widely used optimization workflows is stepwise model simplification. For example, starting from a full or saturated model, the least significant variable ($p > 0.05$) is excluded until the final minimal adequate model is attained ['backward simplification'; Crawley (2007); Forstmeier & Schielzeth (2011); Crawley (2015)]. Each model simplification step will be justified with certain metrics (e.g., SSE, AIC, or BIC) (Henley et al., 2020). Given the number of models, variables of interest, and their data characteristics, this task can be extraordinarily time consuming. Currently, only AIC/BIC-based automated simplification is available (e.g., 'stepAIC,' Venables & Ripley, 2002). Nevertheless, model simplification continues to be a rather manual process [on Google Scholar, only ~5,000 "stepAIC" entries despite ~5,000,000 "linear regression model" studies (0.1%); e.g., Römermann et al. (2016); Karbstein et al. (2019); Henley et al. (2020); Karbstein et al. (2020); Cai et al. (2023); Li (2023)]. In addition, simplification and other aspects such as data cleaning, model comparison and quality control, and output visualization are not yet automated. An easy-to-use, all-in-one function for the entire modeling process within a single software package is missing.

Our R package LazyModeler addresses these issues by automating variable selection, model optimization, and output illustration and generation. In detail, users will be enabled to automatically remove autocorrelated variables, choose between several types of (non)linear regression models (e.g., LM, GLM, LMER, GLMER, GAM, or NLMER), perform stepwise model simplification, check model quality, plot coefficient estimates and relationships, and generate the output of the final model.

Overview and major functions

LazyModeler automates all necessary steps needed for the use of (non)linear regression models. It comprises three major functions that are included within the main function `optimize_model()`.

The first major function `remove_autocorrelations()` checks for any autocorrelations ($|r| > 0.7$) (Dormann et al., 2013) given a list of variables sorted by relevance. Automatic removal of these autocorrelations is possible through the use of a function parameter. Removal will follow the order of the list of variables, ensuring that the user's expertise on the importance of features is respected. A named list is returned with a) a vector containing all removed predictors, and b) a data frame listing autocorrelations and information on deleted variables.

The main function provides the model formula to the second major function `simplify_model()`. If autocorrelations were detected, the formula is updated accordingly. The regression model is then calculated. Options for the models are: `stats::lm()`, `stats::glm()`, `lmerTest::lmer()`, `lme4::glmer()`, `mgcv::gam()`, `nlme::nlme()`, or `stats::nls()`, with all possible distributions of the response variable being allowed. Stepwise backward simplification or forward model selection takes place using an iterative process where each time the metric(s) specified by the user are applied on the model to check whether further simplification or selection is needed. Main variables are kept when they are involved in interactions. Options for the metrics are: `stats::anova()`, `nlme::anova.lme()`, `mgcv::anova.gam()`, `stats::AIC()`, `MuMIn::AICc()`, or `stats::BIC()`. The final model is returned to the main function alongside its metadata as well as simplification history if requested by the user.

Using the third major function `plot_model_features()`, the final model then undergoes multiple visualization steps. Plots to assess model quality are created using the standard plot

function available through base R, or model check included in the performance R package (Lüdecke et al., 2021). Furthermore, the script produces regression, box, or violin plots for each numerical or categorical coefficient as well as plots depicting effects sizes and estimates. All generated plots are returned to the user within a named list. The main function additionally returns the output of both the model simplification or selection and autocorrelation functions as well as the summary of the final model.

LazyModeler makes use of the R package corplot (Wei & Simko, 2021) to calculate correlations between variables, lme4 (Bates et al., 2024), lmerTest (Kuznetsova et al., 2017), mgcv (Wood, 2011), and nlme (Pinheiro et al., 2025) for regression modeling and ANOVA testing, tidyverse (Wickham et al., 2019) for data handling, and MuMIn (Bartoń, 2024) for calculation of AICc scores. For generation of plots visualizing regression, effect size, and estimates, the script further leverages tidyverse and color palettes included in the colorspace (Zeileis et al., 2020) and viridis (Garnier et al., 2024) R packages.

Example

To get to know the package and its main function `optimize_model()`, we provide a testing dataset. This dataset contains information on European plant specimens, including geolocation, habitat factors, and ploidy levels (= the number of complete chromosome sets in a cell's nucleus, ~ genome size) (Karbstein et al., 2021). The corresponding data types are numeric (floats and integers) as well as categorical, offering a range of combinations for the user to test.

A common usage requires an input data frame as well as a starting term for the model. The term needs to be provided as a `quote()` and can encompass transformed variables and interactions (note: we currently only allow for 2-way interactions). If the user wants to check for autocorrelations, they can provide a list of coefficients (in the form of data frame columns) that must be sorted by their relevance in descending order. If an autocorrelation is detected and the parameter `automatic_removal=TRUE` is set, the coefficient further down the list will be removed first.

To calculate the model, the user can provide the type of linear model to calculate (default: "glm"), and the family (default: "gaussian"). The user can also decide on the simplification direction ("forward" for forward selection, "backward" for backward simplification, or "both"). If no simplification is desired, setting `simplification_direction = "backward"` plus `backward_simplify_model = FALSE` yields the original model without simplification.

The following example generates and optimizes a generalized linear model using the provided plant dataset and a term that includes plant reproductive, geographic, ecological, and cytogenetic information. The aim of the model is to identify the cytogenetic (ploidy levels [2n, 4n, etc.]), environmental (altitude in meters above sea level, latitude, longitude, WorldClim solar radiation [$\text{kJ m}^{-2} \text{day}^{-1}$], BioClim annual mean temperature [$^{\circ}\text{C}$], and BioClim isothermality [ratio of mean diurnal range to temperature annual range * 100]) factors that determine the production of sexual seeds (ratio of sexually to asexually formed seeds [%]) in the facultatively asexual model plant group *Ranunculus auricomus* (Ranunculaceae). Please see (Karbstein et al., 2021) for more details. The pipeline checks for correlations between the values of the provided data frame columns and removes autocorrelated variables. The autocorrelation-cleaned term is then used for backward simplification of the model. Finally, the information on the coefficients of the simplified model is plotted. To access the final model, the user can navigate to `models_with_info` within the result, then either to forward or backward depending on the chosen selection/simplification procedure, and then to `final_model`. The plots are stored alongside the model within `plots` - these plots cover the result of `performance::check_model()`, as well as the regression, estimate, and effect size plots.

```
# import example data
data(plants)
```

```
# check data structure
str(plants)
summary(plants)

results_example <- optimize_model(
  df = plants,
  term = quote(sexual_seed_prop ~
    altitude +
    latitude_gps_n +
    longitude_gps_e +
    (solar_radiation +
      annual_mean_temperature +
      isothermality)^2 +
    I(isothermality^2) +
    habitat +
    ploidy),
  autocorrelation_cols = c(
    "solar_radiation",
    "annual_mean_temperature",
    "isothermality",
    "altitude",
    "latitude_gps_n",
    "longitude_gps_e"
  ),
  automatic_removal = TRUE,
  autocorrelation_threshold = 0.8,
  correlation_method = "spearman",
  cor_use = "complete.obs",
  model_type = "glm",
  model_family = "quasibinomial",
  evaluation_methods = c("anova"),
  simplification_direction = "backward",
  backward_simplify_model = TRUE,
  omit_na = "overall",
  scale_predictor = TRUE,
  plot_quality_assessment = "performance",
  round_p = 3,
  plot_relationships = TRUE,
  jitter_plots = TRUE,
  plot_type = "violin",
  stat_test = "wilcox",
  trace = TRUE)
```

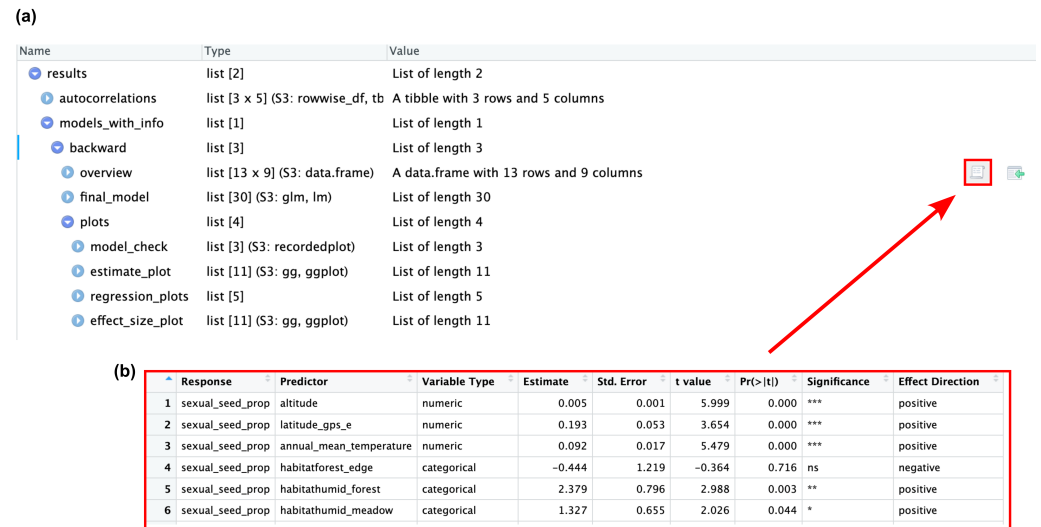


Figure 1: Navigating through the output. For example, (a) simply click on the data frame button highlighted with a red arrow to (b) illustrate the final model output.

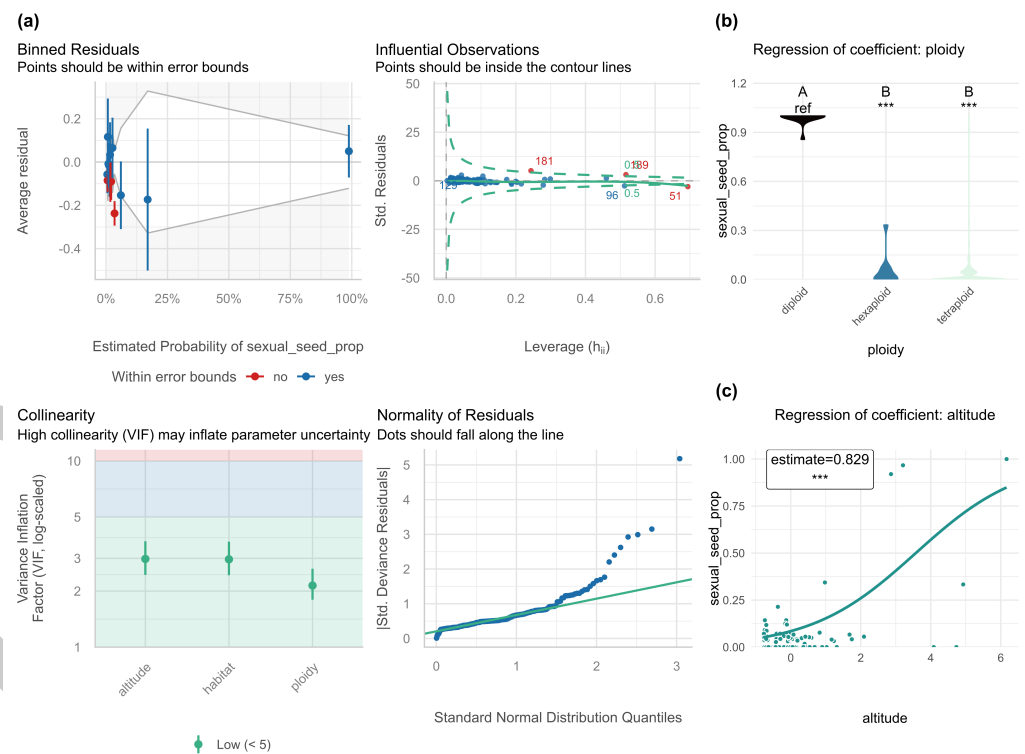


Figure 2: (a) Model quality check provided by `performance::model_check()` and (b,c) exemplary output plots of significant relationships.

Conclusions

In summary, LazyModeler streamlines the process of building, simplifying, and visualizing regression models in R. By automating key steps such as autocorrelation removal, model selection, quality assessment, and output generation, it significantly reduces manual effort. The package is especially valuable for researchers dealing with large and complex datasets who seek a reproducible and statistically sound regression modeling workflow. We anticipate that LazyModeler will serve as a practical and accessible tool for both novice and experienced users in the scientific community.

Important note

The model selection procedures implemented in LazyModeler are provided for convenience and exploratory analysis, and reflect practices recommended in widely used applied statistics literature (Crawley, 2007, 2015). Users should be aware, however, that statistical inference reported from a model chosen in a data-driven way may be anti-conservative (e.g., p-values may appear smaller than they truly are, confidence intervals narrower). This issue is known as post-selection inference (PSI). Specialized methods have been developed to address it (e.g., Lee et al., 2016), but they are not yet broadly applicable across the full range of model classes supported by LazyModeler. We have implemented PSI for (generalized) linear regression models based on the 'selcorr' R package (Cattaneo, 2021), but users are free to use the retained model from LazyModeler for more sophisticated PSI analyses.

Code availability

The code including basic documentation and an exemplary testing dataset will be made available upon publication on [GitHub](#) and on [Comprehensive R Archive Network \(CRAN\)](#).

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References

- Bartoń, K. (2024). *MuMIn: Multi-model inference*. <https://doi.org/10.32614/cran.package.mumin>
- Bates, D., Maechler, M., Bolker, B., & Walker, S. (2024). *lme4 - Linear mixed-effects models using Eigen and S4*. <https://github.com/lme4/lme4/>
- Bauer, M., & Albrecht, H. (2020). Vegetation monitoring in a 100-year-old calcareous grassland reserve in Germany. *Basic and Applied Ecology*, 42, 15–26. <https://doi.org/10.1016/j.baae.2019.11.003>
- Benjamin, A. S., Fernandes, H. L., Tomlinson, T., Ramkumar, P., VerSteeg, C., Chowdhury, R. H., Miller, L. E., & Kording, K. P. (2018). Modern machine learning as a benchmark for fitting neural responses. *Frontiers in Computational Neuroscience*, 12(July), 1–13. <https://doi.org/10.3389/fncom.2018.00056>
- Bzdok, D., & Ioannidis, J. P. A. (2019). Exploration, Inference, and Prediction in Neuroscience and Biomedicine. *Trends in Neurosciences*, 42(4), 251–262. <https://doi.org/10.1016/j.tins.2019.02.001>

- 2178 Cai, L., Kreft, H., Taylor, A., Denelle, P., Schrader, J., Essl, F., Kleunen, M. van, Pergl, J.,
2179 Pyšek, P., Stein, A., Winter, M., Barcelona, J. F., Fuentes, N., Inderjit, Karger, D. N.,
2180 Kartesz, J., Kuprijanov, A., Nishino, M., Nickrent, D., ... Weigelt, P. (2023). Global models
2181 and predictions of plant diversity based on advanced machine learning techniques. *New*
2182 *Phytologist*, 237(4), 1432–1445. <https://doi.org/10.1111/nph.18533>
- 2183 Cattaneo, M. (2021). *Selcorr: Post-selection inference for generalized linear models*. <https://CRAN.R-project.org/package=selcorr>
- 2184
- 2185 Crawley, M. J. (2007). *The R Book* (p. 942). John Wiley & Sons, Ltd. <https://doi.org/10.1002/9780470515075>
- 2186
- 2187 Crawley, M. J. (2015). *Statistics: an introduction using R* (sec. ed., p. 339). John Wiley &
2188 Sons. ISBN: 1118448960
- 2189 Dormann, C. F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., Marquéz, J. R.
2190 G., Gruber, B., Lafourcade, B., Leitão, P. J., Münkemüller, T., McClean, C., Osborne, P.
2191 E., Reineking, B., Schröder, B., Skidmore, A. K., Zurell, D., & Lautenbach, S. (2013).
2192 Collinearity: a review of methods to deal with it and a simulation study evaluating their
2193 performance. *Ecography*, 36(1), 27–46. <https://doi.org/10.1111/j.1600-0587.2012.07348.x>
- 2194 x
- 2195 Forstmeier, W., & Schielzeth, H. (2011). Cryptic multiple hypotheses testing in linear models:
2196 overestimated effect sizes and the winner's curse. *Behavioral Ecology and Sociobiology*,
2197 65(1), 47–55. <https://doi.org/10.1007/s00265-010-1038-5>
- 2198 Garnier, Simon, Ross, Noam, Rudis, Robert, Camargo, Pedro, A., Sciaini, Marco, Scherer,
2199 & Cédric. (2024). *viridis(Lite) - colorblind-friendly color maps for r*. <https://doi.org/10.5281/zenodo.4679423>
- 2200
- 2201 Hastie, T. (2023). *gam: Generalized Additive Models*. <https://cran.r-project.org/web/packages/gam/index.html>
- 2202
- 2203 Henley, S. S., Golden, R. M., & Kashner, T. M. (2020). Statistical modeling methods:
2204 challenges and strategies. *Biostatistics & Epidemiology*, 4(1), 105–139. <https://doi.org/10.1080/24709360.2019.1618653>
- 2205
- 2206 Karbstein, K., Prinz, K., Hellwig, F., & Römermann, C. (2020). Plant intraspecific functional
2207 trait variation is related to within-habitat heterogeneity and genetic diversity in *Trifolium*
2208 *montanum* L. *Ecology and Evolution*, 10(11), 5015–5033. <https://doi.org/10.1002/ece3.6255>
- 2209
- 2210 Karbstein, K., Römermann, C., Hellwig, F., & Prinz, K. (2023). Population size affected
2211 by environmental variability impacts genetics, traits, and plant performance in *Trifolium*
2212 *montanum* L. *Ecology and Evolution*, 13(8), 1–19. <https://doi.org/10.1002/ece3.10376>
- 2213 Karbstein, K., Tomasello, S., Hodač, L., Lorberg, E., Daubert, M., & Hörandl, E. (2021).
2214 Moving beyond assumptions: Polyploidy and environmental effects explain a geographical
2215 parthenogenesis scenario in European plants. *Molecular Ecology*, 30(11), 2659–2675.
2216 <https://doi.org/10.1111/mec.15919>
- 2217 Karbstein, K., Tomasello, S., & Prinz, K. (2019). Desert-like badlands and surrounding
2218 (semi-)dry grasslands of Central Germany promote small-scale phenotypic and genetic
2219 differentiation in *Thymus praecox*. *Ecology and Evolution*, 9(24), 14066–14084. <https://doi.org/10.1002/ece3.5844>
- 2220
- 2221 Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests
2222 in linear mixed effects models. *Journal of Statistical Software*, 82(13), 1–26. <https://doi.org/10.18637/jss.v082.i13>
- 2223
- 2224 Lee, J. D., Sun, D. L., Sun, Y., & Taylor, J. E. (2016). Exact post-selection inference, with
2225 application to the lasso. *Annals of Statistics*, 44(3), 907–927. <https://doi.org/10.1214/15->

226 AOS1371

- 227 Li, J. (2023). Overview of high dimensional linear regression models. *Theoretical and Natural*
 228 *Science*, 5(1), 656–661. <https://doi.org/10.54254/2753-8818/5/20230427>
- 229 Liaw, K., Khomik, M., & Arain, M. A. (2021). Explaining the shortcomings of log-transforming
 230 the dependent variable in regression models and recommending a better alternative:
 231 Evidence from soil CO₂ emission studies. *Journal of Geophysical Research: Biogeosciences*,
 232 126(5), 1–18. <https://doi.org/10.1029/2021JG006238>
- 233 Lüdecke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., & Makowski, D. (2021). Performance:
 234 An r package for assessment, comparison and testing of statistical models. *Journal of Open*
 235 *Source Software*, 6(60), 3139. <https://doi.org/10.21105/joss.03139>
- 236 Pinheiro, J., Bates, D., & R Core Team. (2025). *Nlme: Linear and nonlinear mixed effects*
 237 *models*. <https://doi.org/10.32614/CRAN.package.nlme>
- 238 R Core Team. (2026). *R: a language and environment for statistical computing*. R Foundation
 239 for Statistical Computing. <http://www.r-project.org/>
- 240 Römermann, C., Bucher, S. F., Hahn, M., & Bernhardt-Römermann, M. (2016). Plant
 241 functional traits – fixed facts or variable depending on the season? *Folia Geobotanica*,
 242 51(2), 143–159. <https://doi.org/10.1007/s12224-016-9250-3>
- 243 Schielzeth, H. (2010). Simple means to improve the interpretability of regression coefficients.
 244 *Methods in Ecology and Evolution*, 1(2), 103–113. <https://doi.org/10.1111/j.2041-210X.2010.00012.x>
- 245
- 246 Schielzeth, H., Dingemanse, N. J., Nakagawa, S., Westneat, D. F., Alaguela, H., Teplitsky, C.,
 247 Réale, D., Dochtermann, N. A., Garamszegi, L. Z., & Araya-Ajoy, Y. G. (2020). Robustness
 248 of linear mixed-effects models to violations of distributional assumptions. *Methods in*
 249 *Ecology and Evolution*, 11(9), 1141–1152. <https://doi.org/10.1111/2041-210X.13434>
- 250 Schulz, M.-A., Yeo, B. T. T., Vogelstein, J. T., Mourao-Miranada, J., Kather, J. N., Kording,
 251 K., Richards, B., & Bzdok, D. (2020). Different scaling of linear models and deep learning
 252 in UKBiobank brain images versus machine-learning datasets. *Nature Communications*,
 253 11(1), 4238. <https://doi.org/10.1038/s41467-020-18037-z>
- 254 Venables, W. N., & Ripley, B. D. (2002). *Modern Applied Statistics with S* (Fourth). Springer.
 255 ISBN: 0-387-95457-0
- 256 Wei, T., & Simko, V. (2021). *R package 'corrplot': Visualization of a correlation matrix*.
 257 <https://github.com/taiyun/corrplot>
- 258 Wicke, S., Müller, K. F., DePamphilis, C. W., Quandt, D., Bellot, S., & Schneeweiss, G. M.
 259 (2016). Mechanistic model of evolutionary rate variation en route to a nonphotosynthetic
 260 lifestyle in plants. *Proceedings of the National Academy of Sciences of the United States*
 261 *of America*, 113(32), 9045–9050. <https://doi.org/10.1073/pnas.1607576113>
- 262 Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Golemund,
 263 G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M.,
 264 Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., ... Yutani, H. (2019). Welcome
 265 to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>
- 266
- 267 Wood, S. N. (2011). Fast stable restricted maximum likelihood and marginal likelihood
 268 estimation of semiparametric generalized linear models. *Journal of the Royal Statistical*
 269 *Society (B)*, 73(1), 3–36. <https://doi.org/10.1111/j.1467-9868.2010.00749.x>
- 270 Zeileis, A., Fisher, J. C., Hornik, K., Ihaka, R., McWhite, C. D., Murrell, P., Stauffer, R.,
 271 & Wilke, C. O. (2020). colorspace: A toolbox for manipulating and assessing colors and
 272 palettes. *Journal of Statistical Software*, 96(1), 1–49. <https://doi.org/10.18637/jss.v096>

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